

Tollan Berhanu

Software Engineer | M.S. in CS

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EDUCATION

MSc. Computer Science

University of Idaho

Aug 2024 – Dec 2026

Idaho Falls, ID, USA

GPA: 4.00 / 4.00

Expected Graduation: December 2026

- **Relevant Coursework:** Data Science, Natural Language Processing, Deep Learning, Advanced Networking & Security, Adversarial Machine Learning, Computer Science Research Methods.

BSc. Software Engineering

Jimma University

Oct 2018 – Aug 2022

Jimma, Ethiopia

GPA: 3.97 / 4.00

- **Relevant Coursework:** Data Structures & Algorithms, System Design, Simulation & Modeling, Object-Oriented Programming, Distributed Systems, Artificial Intelligence.
- **Senior Capstone Project:** Automated Exam Scoring using Optical Character Recognition (OCR)
- **Honors:** Awarded a **Gold Medal** for achieving the highest GPA from my department.

WORK EXPERIENCE

Uber Technologies, Inc.

Software Engineering Intern

May 2026 – Aug 2026

Sunnyvale, CA, USA

- Built an **AI agent evaluation pipeline** in **Arize** that traces the agent's execution path — skill invocations, tool calls, sub-agent interactions — and evaluates responses against a ground truth. Used this data to iteratively improve on-call support agents within Uber, raising response accuracy by **70%** and reducing on-call engineer workload by **50%**.
- Built a scheduled **continuous documentation pipeline** that extracts Q&A from Slack support channels, identifies gaps in internal documentation, and drafts updates to Confluence pages for engineers to review and publish, keeping agent knowledge up-to-date with new feature rollouts and reducing repeat questions for human on-calls by **80%**.

Idaho National Laboratory

AI/ML Intern

Nov 2025 – Apr 2026

Idaho Falls, ID, USA

- Developed **FORESIGHT**, a **multi-agent platform** built with **LangGraph** for analyzing and forecasting disruptions in critical minerals supply chains. **FORESIGHT** was featured on the **U.S. Department of Energy (DOE) Genesis Mission** as an aligned national laboratory action.
- Co-authored **two peer-reviewed research papers** on **FORESIGHT** — a Conference Paper submitted to **IEEE - Industrial Electronics Society (IES)** and a Journal submitted to **MDPI - Machine Learning and Knowledge Extraction**.

SingularityNET

Software Engineer

Feb 2023 – Jul 2024

AMS, Netherlands

- Led a team of **five** to create a pipeline for mapping open-source biomedical datasets into **Neo4j** Knowledge Graphs using **Llama 2**, serving as long-term memory for LLMs, increasing user engagement by **60%** over the legacy chat bot.
- Built an open-source **semantic search** plugin for the Mattermost messaging platform using **RAG** methods. Improved search relevance by **30%** and supported **10,000+** monthly queries.
- Contributed to developing and launching the **Cogito stablecoin**, which achieved a market cap of **\$2.96 million** in 2023. Created smart contracts for minting and exchanging stablecoins on the **Cardano blockchain**.

iCog Labs

Machine Learning Engineer Intern

Oct 2022 – Jan 2023

Addis Ababa, Ethiopia

- Analyzed gene expression datasets from leukemia patients to train a **machine learning** classification model, achieving over **90%** accuracy in identifying different types of leukemia.

SELECTED PROJECTS

FORESIGHT — CMM Supply Chain Analysis and Price Forecasting Platform

Internship Project — Idaho National Laboratory

Nov 2025 – Apr 2026

An AI-enabled platform for forecasting disruptions in critical minerals and materials (CMM) supply chains. Users define **custom knowledge graph schemas** and populate them from their own documents to model supply chain dependencies and **forecast price fluctuations**. **FORESIGHT** is estimated to reduce DOE managers' landscape review workload by **~90%**.

- **Tools Used:** Next.js, FastAPI, Neo4j, Postgres, Qdrant, Docker

AE3GIS—An Agile Emulated Educational Environment for Guided Industrial Security

Training

Aug 2025 – Mar 2026

Graduate Research Assistantship — University of Idaho

- An **open-source platform** for ICS cybersecurity training that automates the creation and orchestration of virtual industrial control system environments using GNS3 and containerized IT/OT components. Built an **agentic AI layer** using custom **MCP servers** to automate scenario creation and orchestration. This work was **published** to the **MDPI - Future Internet** journal.
- **Tools Used:** React, FastAPI, MCP Python SDK, GNS3, ContainerLab, Docker

TECHNICAL SKILLS

Gen AI Engineering — Model Context Protocol (MCP), Parameter Efficient Fine-Tuning (PEFT), Retrieval-Augmented Generation

(RAG), LangGraph, Arize • **ML Tools** — Keras, TensorFlow, PyTorch, Scikit-Learn, Matplotlib • **Databases** — MySQL, PostgreSQL,

Neo4j, MongoDB, ChromaDB, Qdrant • **Programming Languages** — Python, JavaScript, Java, Haskell • **Web Frameworks** —

React, Next.js, Electron, Node.js, Flask, Django, FastAPI • **DevOps** — Google Cloud Platform, AWS, CI/CD, Docker, Git